Narges Rezaie

Data Scientist, Bioinformatics Engineer, Computational Biologist, Data Analyst, AI/ML Engineer

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Summary

Bioinformatics Engineer & **Ph.D. Computational Biologist** (**UC Irvine**) with **7+ years** of experience. Expert in building **scalable pipelines** for **long/short-read sequencing** (**ONT, PacBio, Illumina**), **bacterial genomics**, and **single-cell multi-omics**. Specialist in **Nextflow, Snakemake,** and **AWS cloud-native** deployments. Pioneering **LLM-based agentic systems** to automate **bioinformatics workflows** and software tooling.

Work Experience

Bioinformatics Pipeline Engineer — **CA, USA**
Plasmidsaurus — *Dec 2024 – Present*

- **Own the end-to-end development** and **productionization** of bioinformatics pipelines, serving as the **primary lead** for **bacterial production workflows** and RNA-seq product support.
- **Architect scalable workflow systems** using **Nextflow** and **Snakemake** across **AWS Batch** and HPC environments, ensuring **reproducible results** for high-throughput sequencing data.
- **Lead the launch** of **small variant calling** services and **driving development** of **Illumina metagenomics** through pilot phase; **benchmarked and integrated** advanced tools for **taxonomic profiling** and **host-read removal** to optimize **turnaround time** and result accuracy.
- **Optimize pipeline performance** by resolving complex **biological, sequencing, and infrastructure edge cases** from initial data ingestion to the final customer deliverable.
- **Drive platform evolution** as a **full-stack contributor**, developing **backend logic** and **frontend features** that enhance **data visualization** and the seamless delivery of genomic results.
- **Pioneer the integration** of **LLM and Agent-based systems** to automate **bioinformatics workflows**, streamline complex data analysis, and elevate **internal developer tooling**.
- **Collaborate cross-functionally** with scientists, engineers, and product teams to **translate novel computational methods** into **robust, customer-facing** bioinformatics products.

Research Experience

Postdoctoral fellow — **CA, USA**
Mortazavi Lab, University of California, Irvine — *March 2024 – Jan 2025*

- Developed, applied, and optimized **bioinformatic algorithms**, tools, and **workflows** to analyze **NGS datasets** including **short- and long-read RNA-seq** at **bulk and single-cell resolution**, as well as **Multiome**, **Perturb-seq**, and **DNA methylation** data.
- Characterized the impact of **genomic variants**, **regulatory elements**, and genes on molecular and cellular phenotypes by analyzing naturally occurring or designed **genomic perturbations** across various **cellular models** (Publication).
- Identified **active regulatory elements** and genes for individual **cell types and states** by analyzing **single-cell datasets**.
- **Gene program evaluation** (**GitHub**): Implemented a **flexible framework** to evaluate the plausibility of **gene programs** inferred from single-cell genomic data (**scRNA-seq, scATAC-seq, Multiome, and Perturb-seq**) using computational methods.

Graduate Researcher — **CA, USA**
Mortazavi Lab, University of California, Irvine — *Sept 2019 – March 2024*

- **Model-AD** (**GitHub**): Analyzed diverse genomics datasets (**bulk, single-cell, and single-nucleus**) to characterize **next-generation animal models** for **Alzheimer's disease** and develop **predictive models** for therapeutic discovery. Identified **gene and transcript expression changes** in Model-AD mice relative to wild-type mice and **human AD patients**. Implemented **standardized data acquisition** to Synapse.
- **IGVF**: Investigated how **genomic variation** affects genome function and phenotype across tissues using **single-cell data**.

- **ENCODE4**: Integrated and interpreted **single-cell omics data** to profile **full-length transcriptomes** in humans and mice.
- **Gene Regulatory Networks**: Built models of **gene regulatory networks** using **bulk and single-cell RNA-seq and ATAC-seq** data in humans, rodents, and other vertebrates to study how **developmental logic** is encoded in the genome.
- **PyWGCNA (GitHub)**: Developed a **Python package** for **weighted correlation network analysis (WGCNA)**, enabling **comparison of networks across datasets** and against external gene lists using **Fisher's exact test** to assess module functional enrichment.
- **Topyfic (GitHub)**: Developed a **Python package** to identify **gene programs (topics)** using a robust **unsupervised machine learning** approach based on **Latent Dirichlet Allocation (LDA)**, and annotated gene programs using internal and external resources.
- **Benchmarked doublet detection** methods on **single-nucleus RNA sequencing (snRNA-seq)** datasets.
- Collaborated with **experimental scientists** to support research goals and mentored **undergraduate students** in programming and **bioinformatics pipelines**.

Education

PhD in Mathematical, Computational, and Systems Biology — **USA**
University of California, Irvine — *2019–2024*

- **Thesis**: Computational approach to characterize gene dynamics using bulk and single-nucleus RNA sequencing to study Alzheimer's disease. **Supervisor**: Dr. Ali Mortazavi. **GPA**: 3.98/4.0

B.S. in Information Technology in Computer Engineering — **Iran**
Sharif University of Technology — *2014–2018*

Skills

- **Bioinformatics & Computational Biology:** Single-cell/Single-nucleus RNA-seq, Multi-omics integration (scATAC-seq), Spatial Omics, Genome assembly (Hifiasm, Flye), Variant calling (Clair3), Metagenomics, and Benchmarking.
- **Pipeline Engineering & Cloud:** Production-grade **Nextflow (DSL2)** and **Snakemake** development; **AWS** ecosystem (**Batch, S3, EC2, Lambda**); Containerization (**Docker**); **CI/CD** (GitHub Actions); High-Performance Computing (**SLURM**).
- **Software Development: Python (Expert)**, R, Bash, SQL, C/C++; Web frameworks (**FastAPI**, **Pydantic**); Software testing (**Pytest**); Collaborative development (**Git**).
- **AI & Machine Learning: LLM-based Agentic Workflows**, Vector Databases, PyTorch, Scikit-learn, Unsupervised Learning (LDA, WGCNA), and Statistical Modeling.
- **Data Science Stack: Polars** (high-performance dataframes), Pandas, NumPy, SciPy, **AnnData**, Biopython, Matplotlib.
- **Domain Tools:** Seurat, Scanpy, Cellbender, Kallisto, Samtools, Bedtools, DEseq2/EdgeR.
- **Developed Packages: PyWGCNA**, **Topyfic**.

Selected Publications

- Identification of robust cellular programs using reproducible LDA that impact sex-specific disease progression in different genotypes of a mouse model of AD. **N Rezaie**, E Rebboah, et al. *Briefings in Bioinformatics* (2025).
- The ENCODE mouse postnatal developmental time course identifies regulatory programs of cell types and cell states. E Rebboah, **N Rezaie**, et al. *Cell Genomics* (2025).
- Long-read sequencing transcriptome quantification with lr-kallisto. RK Loving, ..., **N Rezaie**, ..., L Pachter. *PLoS Computational Biology* 21(12): e1013692 (2025).
- BIN1^{K358R} suppresses glial response to plaques in mouse model of Alzheimer's disease. LF Garcia-Agudo, ..., **N Rezaie**, ..., KN Green. *Alzheimer's & Dementia* (2025).
- The Abca7^{V1613M} variant reduces Aβ generation, plaque load, and neuronal damage. CA Butler, ..., **N Rezaie**, ..., GR MacGregor, KN Green. *Alzheimer's & Dementia* (2025).
- The ENCODE4 long-read RNA-seq collection reveals distinct classes of transcript structure diversity. F Reese, ..., **N Rezaie**, et al. *Nature Methods* (2024).
- Deciphering the impact of genomic variation on function. IGVF Consortium. *Nature* (2024).

- A Trem2^{R47H} mouse model without cryptic splicing drives age-and disease-dependent tissue damage and synaptic loss in response to plaques. KM Tran, ..., **N Rezaie**, et al. *Molecular Neurodegeneration* (2023).
- PyWGCNA: A Python package for weighted gene co-expression network analysis. **N Rezaie**, F Reese, A Mortazavi. *Bioinformatics* (2023).
- Somatic point mutations are enriched in non-coding RNAs with possible regulatory function in breast cancer. **N Rezaie**, ..., H Alinejad-Rokny. *Communications Biology* (2022).

Selected Talks and Posters

- Alzheimer's Association International Conference 2024. *Poster*: "5xFAD/Abi3^{S209F} mice display distinct effects on microglia dynamics."
- PARSE Biosciences 2023. *Invited Speaker*: "Identifying Distinct Cellular Programs Using Topyfic."
- Network Biology 2023 (CHSL). *Poster*: "Topyfic identified robust cellular programs using reproducible LDA."
- Alzheimer's & Dementia 2022 (AAIC). *Poster*: "Bulk and single-cell gene expression network analysis using PyWGCNA."
- Neuroscience 2020 (SFN). *Poster*: "Bulk and single-nucleus analysis of the 3xTgAD mouse model."

References

Available upon request.